MATE L11

Diffusion in defect & material imperfection

Defect-mediated diffusion (intrinsic) in last lecture

$$D_A = \frac{Z\langle r^2\rangle \nu}{6} \exp\left(\frac{S_v^f + S_v^m}{k_B}\right) \exp\left(-\frac{H_v^f + H_v^m}{k_B T}\right) f$$

Dependent on both

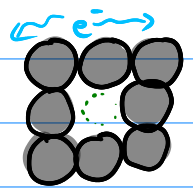
① Vacancy formation S_v^f H_v^f $(H_v^f > 0)$

② vacancy migration barrier S_v^m H_v^m

$\langle r^2\rangle$ Z ν are determined by structural factors

Ionic crystals, however have more types of defects!

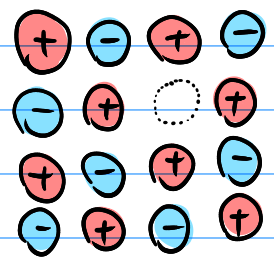
Metal / Alloy



electroneutrality

metal is conducting

Ionic crystal

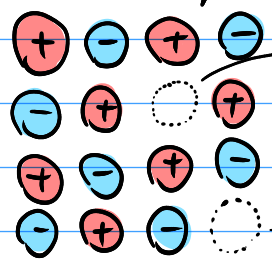


Only creating 1 vacancy → excess charge !

In ionic crystals, defects always come in pairs!

Two types of defects are commonly present.

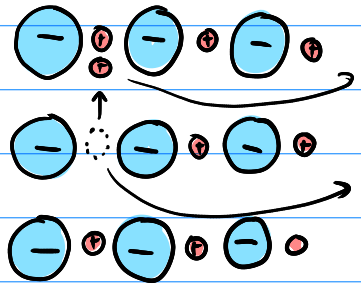
Schottky defect



originally − ⇒ now vacancy ⇒ net charge $\boxed{+1}$

originally + ⇒ now vacancy ⇒ net charge $\boxed{-1}$

Frenkel defect (pair)



originally interstitial site, now ⊕ ⇒ net $\boxed{+1}$

originally ⊕ now vacancy ⇒ net $\boxed{-1}$

How do we write the defect?

Kröger-Vinker (KV) notation

X^{z}_{Y}

z → effective charge $+ \rightarrow \bullet$, $- \rightarrow \prime$

Y → what is replaced?

X ⇓ (what currently at the site?)

AgCl crystal Frenkel defect

V'_{Ag} ⇒ originally Ag, replaced by V, net charge $\boxed{-1}$

$Ag^{\bullet}_{i}$ ⇒ originally interstitial, replaced by Ag, net $\boxed{+1}$

KCl Schottky defect

V_K' $V_{Cl}^{\bullet}$ $\Rightarrow$ K & Cl both replaced by Vacancy

How do we get the ionic-defect-mediated D?

① Calculate concentration $[V_K']$ $[V_{Cl}^{\bullet}]$ (fraction)

② Which specie to diffuse? e.g. $K^+ \longleftrightarrow V_K'$

③ Calculate $D = [V_K'] \cdot D_K$

Equilibrium fraction of defects follow "chemical reaction"

E.g. KCl Schottky defect

$$\text{null} \longrightarrow V_K' + V_{Cl}^{\bullet} \quad \text{reaction } \Delta G_S^f \text{ (Schottky formation)}$$

ΔG_S^f is linked to the equilibrium constant K_S^{eq} (dimensionless)

$$\Delta G_S^f = -k_B T \ln K_{eq}$$

$$K_S^{eq} = [V_K'][V_{Cl}^{\bullet}] = \exp\left(-\frac{\Delta G_S^f}{k_B T}\right)$$

If we started from null $\Rightarrow$ $[V_K']^{eq} = [V_{Cl}^{\bullet}]^{eq} = \sqrt{\exp\left(-\frac{\Delta G_S^f}{k_B T}\right)}$

$$= \exp\left(-\frac{\Delta G_S^f}{2k_B T}\right)$$

Do the same for Frenkel defect

$$\underbrace{Li_{Li}^{x}}_{\text{original site}} \longrightarrow V_{Li}' + Li_i^{\bullet}$$

In a solid $[Li_{Li}^{x}]$ is constant, again $K_F^{eq} = [V_{Li}'] \cdot [Li_i^{\bullet}]$

For Frenkel defect, similarly

$$[V_{Li}'] = [Li_i^{\bullet}] = \exp\left(-\frac{\Delta G_F^f}{2k_BT}\right)$$

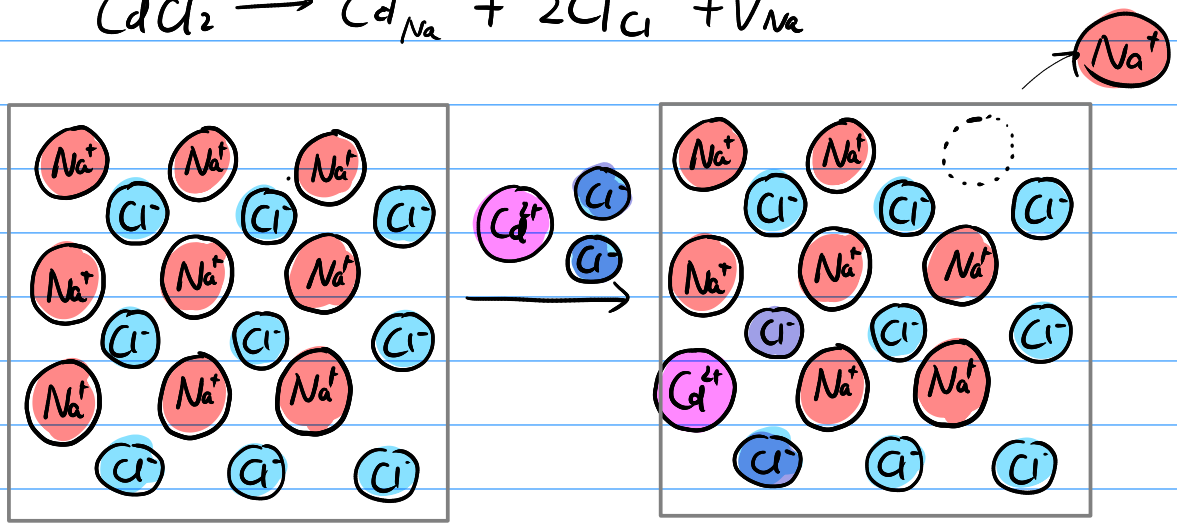
When defects (Schottky or Frenkel alike) are intrinsic their concentration follow thermal activation

The second type of defect is extrinsic, via doping

E.g. KCl crystal. $\begin{cases} \text{Intrinsic defect is Schottky-type} \\ \text{Extrinsic defect: adding } CaCl_2 \end{cases}$

In class, NaCl with $CdCl_2$ dopant

$$CdCl_2 \longrightarrow Cd_{Na}^{\bullet} + 2Cl_{Cl}^{\times} + V_{Na}'$$



The dopant $CdCl_2$ usually has a concentration greater than room temp defect density, so.

$$[V_{Na}']^{ext} \doteq [CdCl_2]$$

If we don't distinguish the sources for defects then overall charge neutrality holds

$$\underset{+1}{[Cd_{Na}^{\bullet}]} + \underset{+1}{[V_{Cl}^{\bullet}]} = \underset{-1}{[V_{Na}']}$$

$$[V_{Na}'] \cdot [V_{Cl}^{\bullet}] = [V_{Na}'] ([V_{Na}'] - [Cd_{Na}^{\bullet}])$$

$$= K_s^{eq} = \exp\left(-\frac{\Delta G_s^f}{k_B T}\right)$$

We can solve $[V_{Na}']$ by a quadratic equation

$$[V_{Na}']^2 - [Cd_{Na}^{\bullet}][V_{Na}'] - \exp\left(-\frac{\Delta G_s^f}{k_B T}\right) = 0$$

$$\left(\text{root} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\right)$$

$$[V_{Na}'] = \frac{1}{2}\left[[Cd_{Na}^{\bullet}] + \sqrt{[Cd_{Na}^{\bullet}]^2 + 4\exp\left(-\frac{\Delta G_s^f}{k_B T}\right)}\right]$$

If exp part is very small compared to $[Cd_{Na}^{\bullet}]$,

$$R.H.S = [Cd_{Na}^{\bullet}] \cdot \sqrt{1 + \left(2 \cdot \frac{\exp\left(-\frac{\Delta G_s^f}{2k_B T}\right)}{[Cd_{Na}]}\right)^2}$$

$y = (1 + x^2)^{\frac{1}{2}}$ Taylor expansion near $x = 0$

$$y = y_0 + \left.\frac{dy}{dx}\right|_{x=0} (x) + \frac{1}{2}\left.\frac{d^2y}{dx^2}\right|_{x=0} (x)^2$$

$$\left.\frac{dy}{dx}\right|_{x=0} = \left.\frac{\frac{1}{2} \cdot 2x}{\sqrt{1+x^2}}\right|_{x=0} = \frac{x}{\sqrt{1+x^2}} = 0$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=0} = \left.\frac{1}{\sqrt{(1+x^2)^3}}\right|_{x=0} = 1 \qquad \text{so } y \simeq 1 + \frac{1}{2}x^2$$

$$R.H.S \doteq [Cd_{Na}^{\bullet}]\left[1 + \frac{1}{2}\left(\frac{2\exp\left(-\frac{\Delta G_s^f}{2k_B T}\right)}{[Cd_{Na}^{\bullet}]}\right)^2\right]$$

In lecture notes, we approximate that

$$[V_{Na}'] \doteq \frac{1}{2}[Cd_{Na}^{\cdot}]\left[1 + \sqrt{1 + 2\left[\frac{\exp(-\frac{\Delta G_S^f}{2k_BT})}{[Cd_{Na}^{\cdot}]}\right]^2}\right]$$

$$= [Cd_{Na}^{\cdot}] + \frac{\exp(-\frac{\Delta G_S^f}{k_BT})}{[Cd_{Na}^{\cdot}]}$$

$[Cd_{Na}^{\cdot}] = [CdCl_2] =$ extrinsic

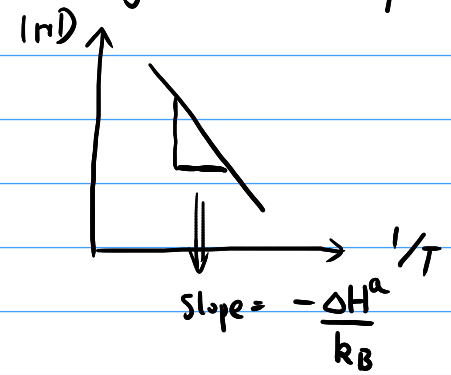
So $[V_{Na}'] \approx [CdCl_2] + \dfrac{\exp(-\frac{\Delta G_S^f}{k_BT})}{[CdCl_2]}$

The other regime is ΔG_S^f is huge, so we will have

$$[V_{Na}'] \doteq \exp(-\frac{\Delta G_S^f}{2k_BT})$$ (Don't use equation above!)

Diffusion in imperfections

Reading Arrhenius plot : Steeper slope => more positive ΔH^a



In metal, one can usually scale by

$\frac{\Delta H^a}{k_B} = \alpha \cdot T_m$ α : factor (1 ~ 30)

T_m : melting point

$D \simeq D_0 \cdot \exp(-\frac{\alpha T_m}{T})$ we can plot T_m/T

So it means α_{xl} (bulk crystal) $> \alpha_L$ (liquid)

$\alpha_{xl} > \alpha_{imp}$

Generally speaking α w.r.t dimension scaling

Pure crystal (α_{xl}) > 0D vacancy (α_v) > 1D dislocation (α_{dis})

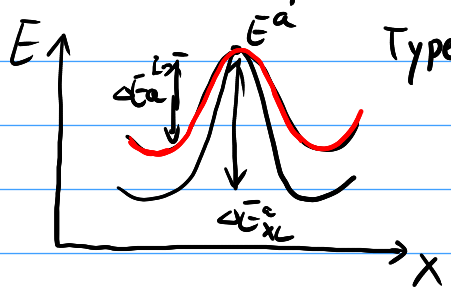
V

2D grain boundary

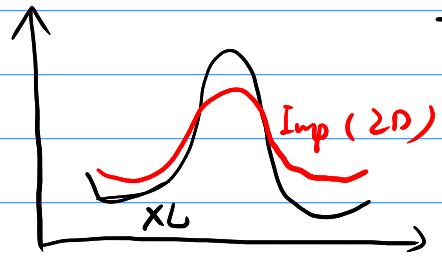
V

2D free surface

Illustration of lower activation energy



Type 1: imperfection at higher E



Type 2: imperfection / surface at low activation barrier (less neighbours)

Harrison ABC model

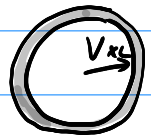
Bulk D^{XL}

GB D^B

Avg grainsize s

Atomic fraction in G.B.

$\eta = \langle \text{atoms in GB} \rangle / \langle \text{total atoms in grain} \rangle$



Surface area $4\pi r^2$

Volume $\frac{4}{3}\pi r^3$

Surface-to-area ratio $= 3 \cdot \frac{1}{r}$

Approximated $\eta = \frac{3\delta}{s}$

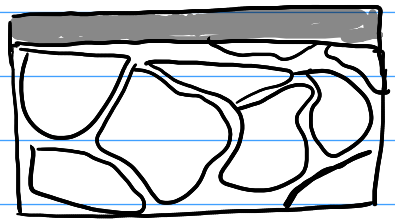
Total diffusion length $D_{eff} \cdot t = D_{XL}(1-\eta)t + D_B \eta t$

If $\eta \ll 1$ $D_{eff} \simeq D_{XL}\overset{\approx 1}{(1-\eta)} + D_B \eta$

$= D_{XL} + D_B \cdot \eta > D_{XL}$

(If $D_{XL} \cdot t \gg s^2$)

ABC model deals with intermediate states



Initially coated top surface, measure diffusion?

λ: interatomic distance

s: grain size

Has diffusion stated in GB, but not into grain?

C region: $D_{XL} t < \lambda^2$ $D_B t > \lambda^2$

B region: $\lambda^2 \simeq D_{XL} t < s^2$ $D_B t > \lambda^2$